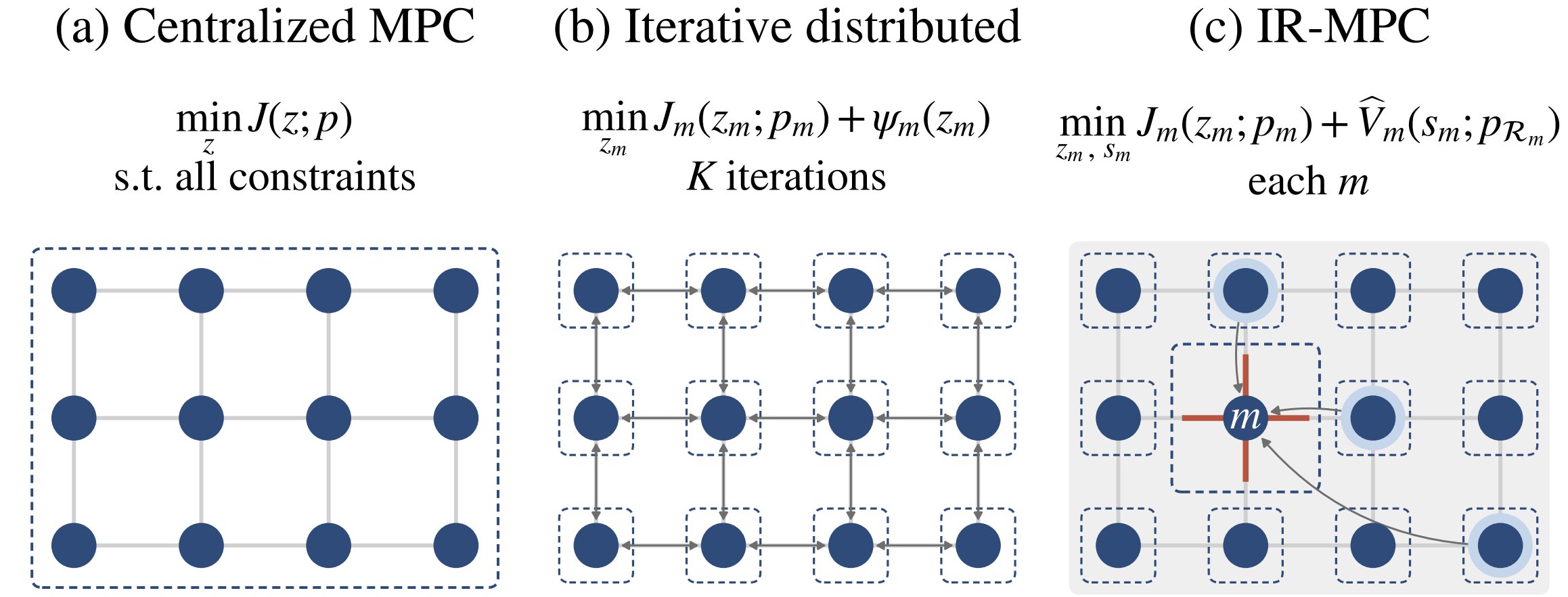
(a) Centralized MPC
$\min_{z} J(z;p)$
s.t. all constraints
(b) Iterative distributed
$\min_{z_m} J_m(z_m;p_m)+\psi_m(z_m)$
K iterations
(c) IR-MPC
$\min_{z_m,\, s_m} J_m(z_m;p_m)+\widehat{V}_m(s_m;p_{\mathcal{R}_m})$
each m
m
decision variables
QP scope
information exchange